IT303: Software Engineering

Real-Time Traffic Sign Recognition System – User Manual

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How to read the screenshots: a **green box** marks the button, field or area you need to click or look at. The small green number on each box matches the numbered step in the text below it.

1. Registration

On the Sign In page click **Create one** (box 5 in Figure 2) to open the registration page.

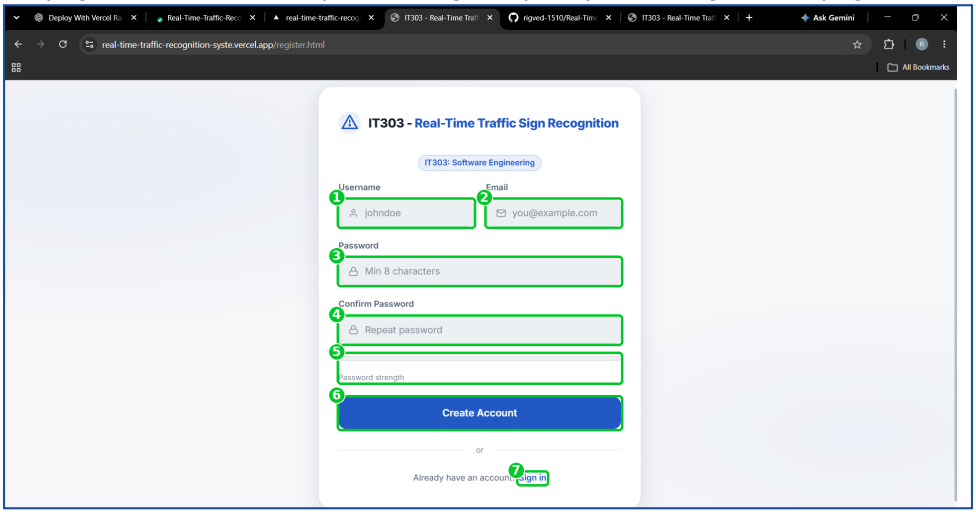


Figure 1: Create Account page

1. **Username (1)** – enter a username (e.g. johndoe).
2. **Email (2)** – enter your email address.
3. **Password (3)** – minimum 8 characters.
4. **Confirm Password (4)** – repeat the password.
5. Watch the **Password strength** bar **(5)** below the fields; use a stronger password if it stays short or red.
6. Click **Create Account (6)**. If you already have an account, click **Sign in (7)**.

2. Sign In

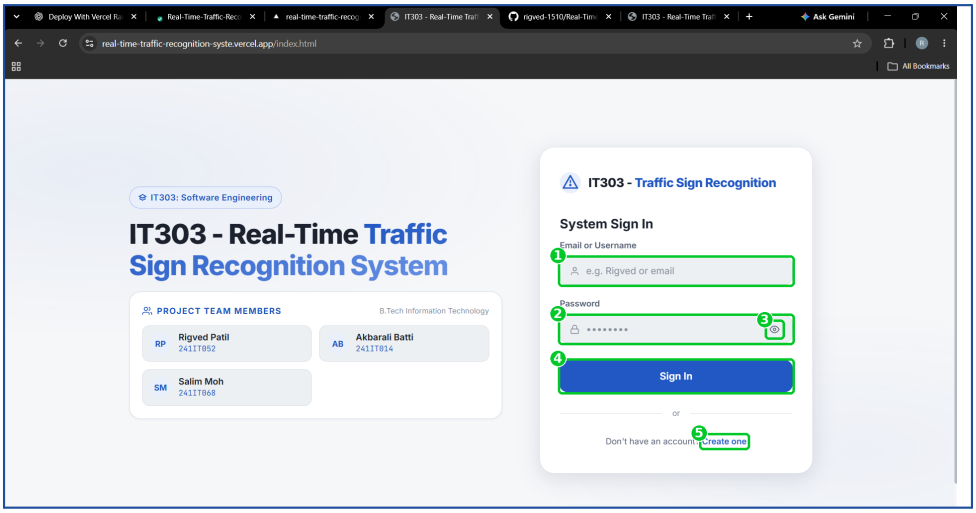


Figure 2: System Sign In page

1. **Email or Username (1)** – type your email or username.
2. **Password (2)** – type your password; click the eye icon **(3)** to show or hide it.
3. Click **Sign In (4)**.
4. New users can click **Create one (5)** to register.

3. Two-Factor Authentication

Your account is protected with a one-time code from an authenticator app. Set it up once on the setup page below.

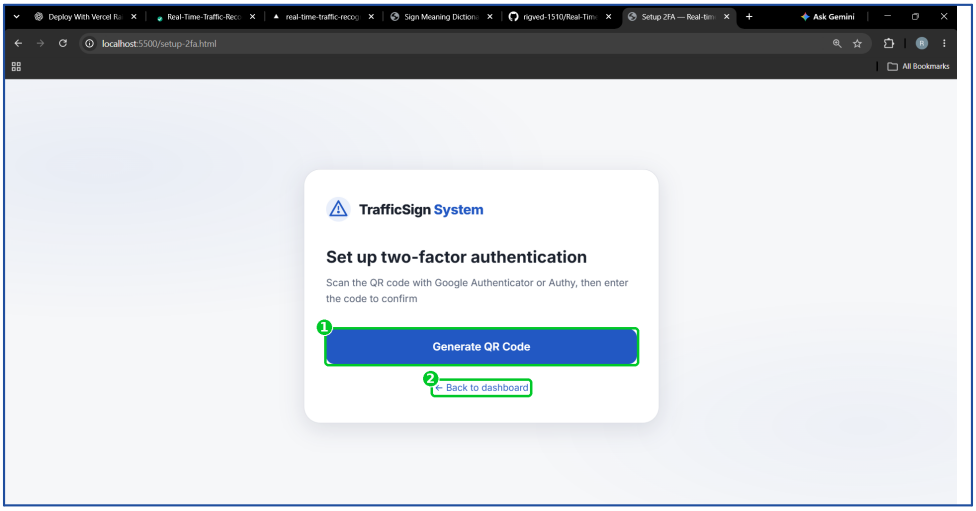


Figure 3: Set up two-factor authentication page

1. Click **Generate QR Code (1)**.
2. Scan the QR code with **Google Authenticator** or **Authy**.
3. Enter the 6-digit code shown in the app to confirm the setup.

- **Back to dashboard (2)** – leaves the setup page and returns to the Dashboard.

On later sign-ins, the Two-Factor Verification page asks for the current code.

1. Open your authenticator app and type the 6-digit code into the six boxes.
2. Click **Verify**.

- **Use a backup code** – use this if you have lost access to your authenticator.
- **Back to login** – returns to the Sign In page.

4. Dashboard

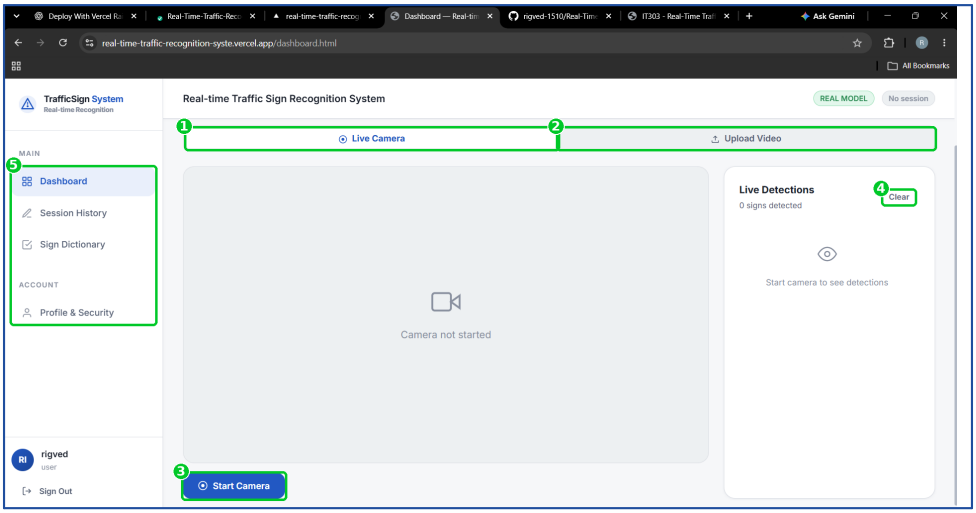


Figure 4: Dashboard (camera not started)

- **Left menu (5)** – Dashboard, Session History, Sign Dictionary and Profile & Security. Your name, role and **Sign Out** are at the bottom.
- **Top bar** – "REAL MODEL" shows the real model is in use; "No session" changes to the session number (e.g. Session #15) when detection runs.
- **Live Camera (1)** / **Upload Video (2)** – tabs to choose the input type.
- **Start Camera (3)** – starts the camera and a new recognition session.
- **Live Detections** – lists detected signs with the count (e.g. "0 signs detected"); **Clear (4)** empties the list.

5. Live Detection

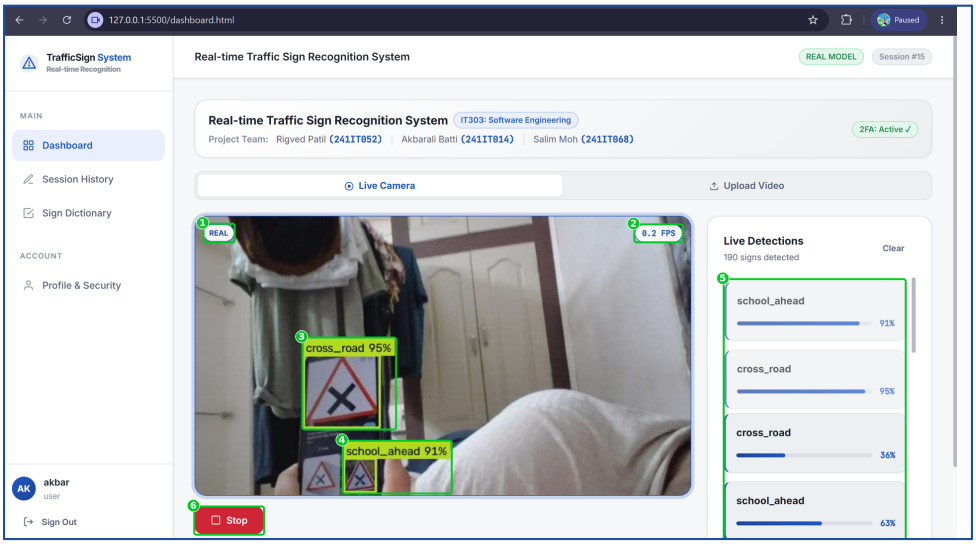


Figure 5: Live detection in progress

1. Select the **Live Camera** tab and click **Start Camera** (Figure 4).
2. Show a traffic sign to the camera. A box with the sign name and confidence is drawn on the video, e.g. *cross_road 95%* **(3)** and *school_ahead 91%* **(4)**.
3. Each detection is added to **Live Detections (5)** with a confidence bar and percentage; the total count is shown (e.g. 190 signs detected). Scroll the list to see older detections.
4. The **REAL** tag **(1)** and the FPS counter **(2)** (e.g. 0.2 FPS) appear on the video; the top bar shows the session number (e.g. Session #15). Higher FPS means smoother detection.

5. Click the red **Stop** button **(6)** to end the session.

6. Session History

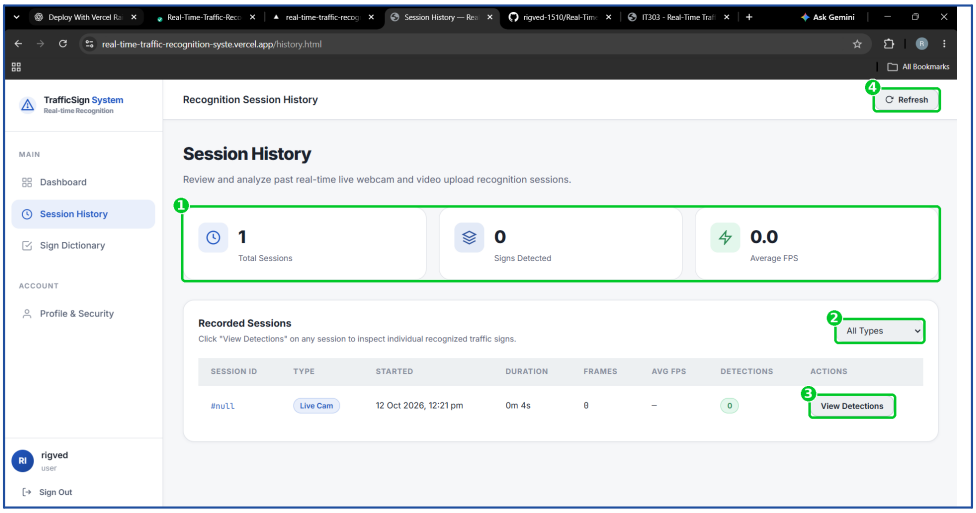


Figure 6: Session History page

- **Summary cards (1)** – Total Sessions, Signs Detected and Average FPS.
- **Recorded Sessions table** – Session ID, Type (Live Cam), Started, Duration (“Active” for a running session), Frames, Avg FPS, Detections and Actions.
- **View Detections (3)** – opens the signs recognised in that session.
- **All Types** drop-down **(2)** filters sessions by type; **Refresh (4)** reloads the data.

7. Sign Dictionary

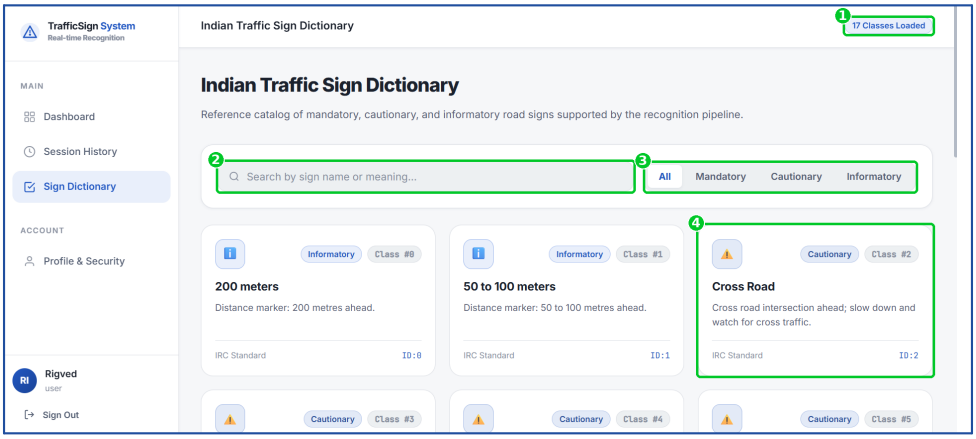


Figure 7: Indian Traffic Sign Dictionary

- The top-right badge **(1)** shows the number of classes loaded (17).
- Type in the **search box (2)** to find a sign by name or meaning, or filter using **All, Mandatory, Cautionary** and **Informatory (3)**.
- Each card **(4)** shows the category, Class number, sign name, meaning, “IRC Standard” and ID (e.g. Cross Road – Class #2).

8. Profile & Security

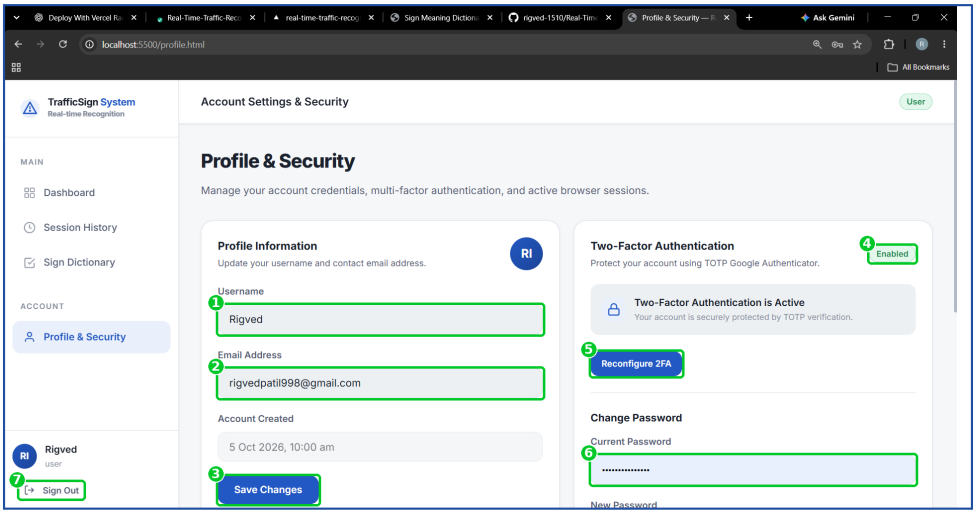


Figure 8: Profile & Security page

- **Profile Information** – edit **Username (1)** and **Email Address (2)**, then click **Save Changes (3)**. Account Created is read-only.
- **Two-Factor Authentication** – the **Enabled** badge **(4)** confirms 2FA is on; click **Reconfigure 2FA (5)** to set it up again.
- **Change Password** – enter your **Current Password (6)**, then scroll down and enter the New Password.
- **Sign Out (7)** (bottom of the left menu) logs you out.